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**Hands-On SIEM Project: Brute Force,
Persistence & FIM Detection with Wazuh —
Including Custom Detection Rule**

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Project Overview

This project involved setting up a Wazuh SIEM in a home lab using two virtual machines — Ubuntu as the Wazuh server and Windows 10 as the monitored agent. I simulated three real-world attack techniques, analyzed the generated alerts in the Wazuh dashboard, and wrote a custom detection rule to extend Wazuh's built-in capabilities.

Disclaimer: All simulations were performed in an isolated virtual environment for educational purposes only. Never attempt these techniques on real devices or systems you do not own.

Lab Environment

Component	Details
Virtualization	Oracle VirtualBox
Wazuh Server	Ubuntu VM — Wazuh Manager + Dashboard
Monitored Agent	Windows 10 VM — Wazuh Agent (Active)

Attack Simulations

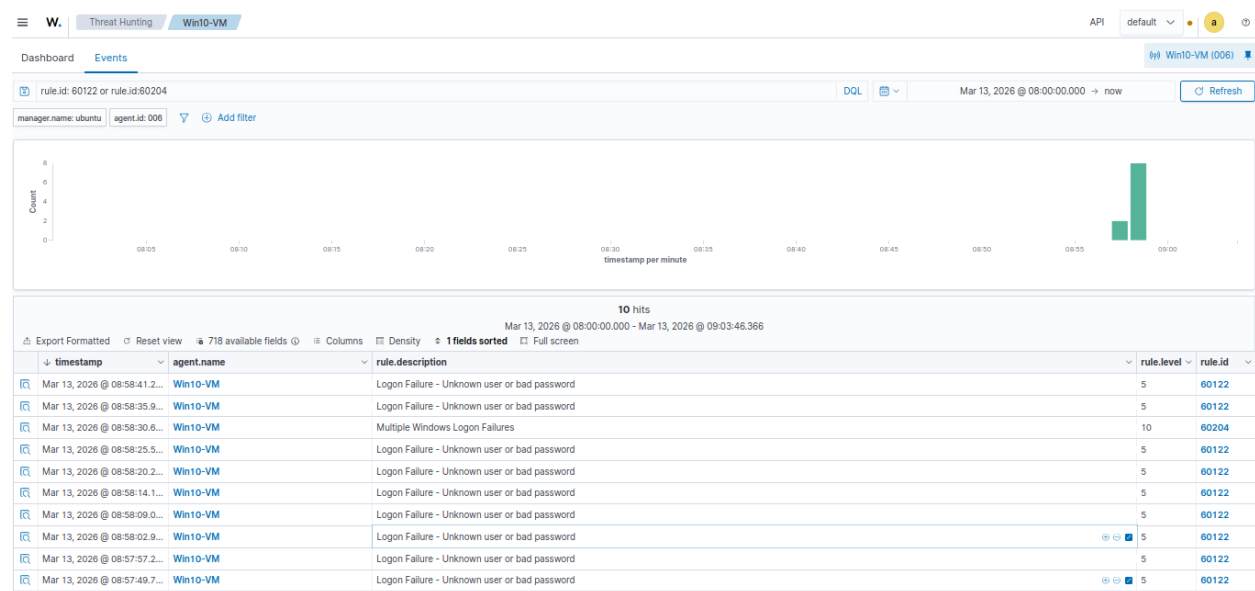
Attack 1 — Brute Force Login

What I did: Ran a failed login command 10 times against my own machine using a fake username and wrong password to simulate a brute force attack.

Command used:

```
PS C:\Windows\system32> net use \\localhost\IPC$ /user:newfake123 abcdef
System error 1326 has occurred.
```

What Wazuh detected:



Attack 2 — Unauthorized Admin Account Creation

What I did: Created a fake user account, gave it administrator privileges, then deleted it — simulating how attackers create backdoor accounts and then cover their tracks.

Commands Used:

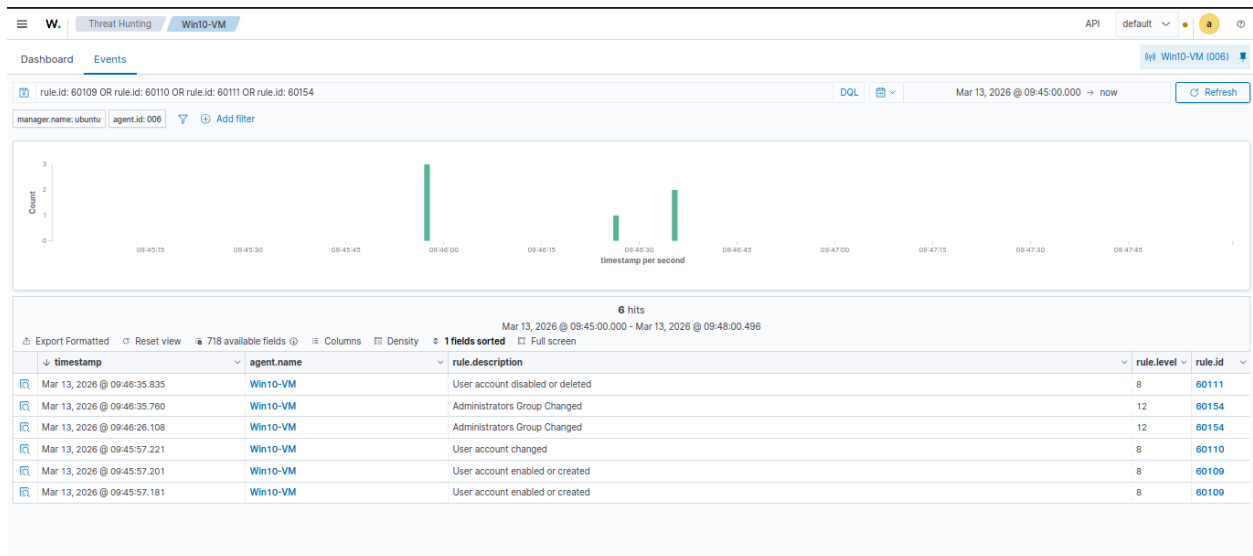
```
Administrator: Windows PowerShell
PS C:\Windows\system32> net user user1234 pass1234 /add
The command completed successfully.

PS C:\Windows\system32> net localgroup administrators user1234 /add
The command completed successfully.

PS C:\Windows\system32> net user user1234 /delete
The command completed successfully.

PS C:\Windows\system32>
```

What Wazuh detected:



What I learned: Windows fires multiple events from a single command. Even though the attacker deleted the account, Wazuh had already logged every step including the admin privilege escalation at Level 12.

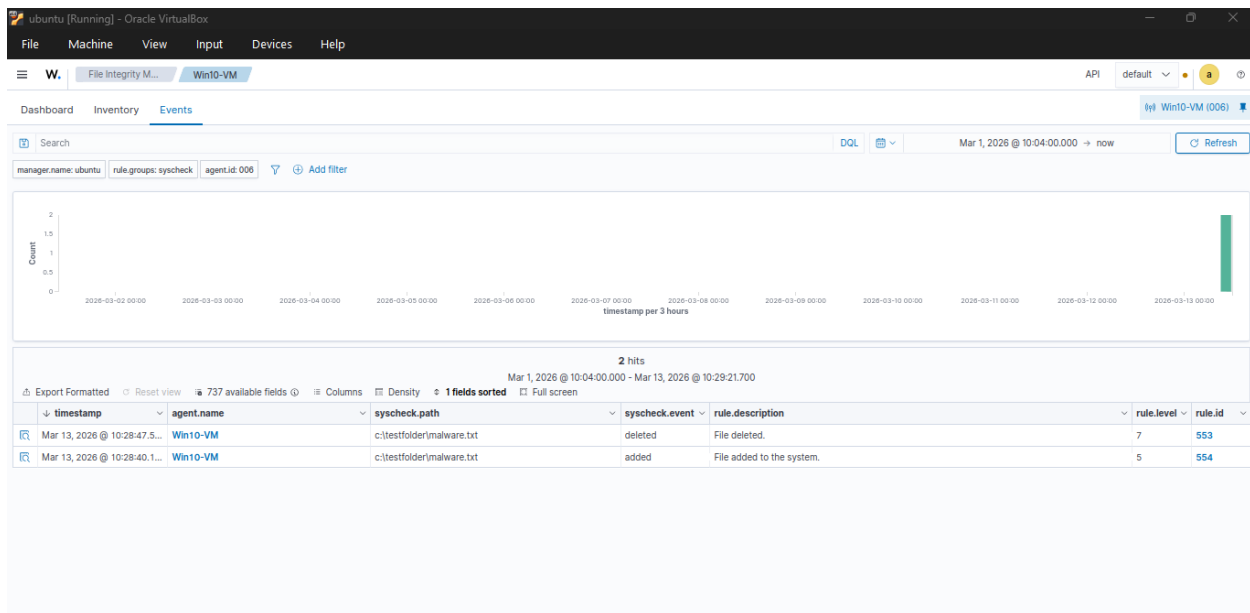
Attack 3 — Malicious File Drop (FIM)

What I did: Configured Wazuh's File Integrity Monitoring (FIM) module to watch a custom folder by editing the Wazuh agent config file on Windows. Then created and deleted a suspicious file to simulate a malware drop.

Commands used:

```
PS C:\Windows\system32> echo "hacker was here" > C:\TestFolder\malware.txt
PS C:\Windows\system32> del C:\TestFolder\malware.txt
```

What Wazuh detected:



Custom Detection Rule

What I did: Wrote my own custom Wazuh detection rule in XML and deployed it to the Wazuh server. The rule builds on top of Wazuh's built-in FIM rule 554 and escalates any file creation in C:\TestFolder to Level 10 with a custom alert message.

File edited:

/var/ossec/etc/rules/local_rules.xml on Ubuntu VM

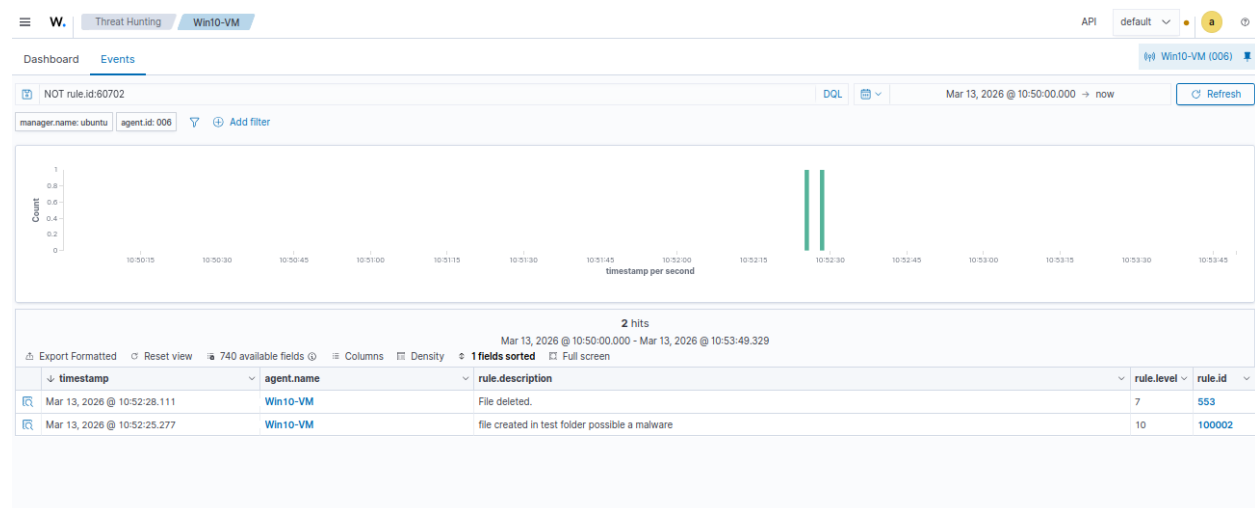
Rule written:

```
<group name="local,siem_project">
  <rule id="100002" level="10">
    <if_sid>554</if_sid>
    <field name="file">C:\\TestFolder</field>
    <description>file created in test folder possible a malware</description>
  </rule>
</group>
```

Commands Used:

```
PS C:\Windows\system32> echo "hacker was here" > C:\TestFolder\malware.txt
PS C:\Windows\system32> del C:\TestFolder\malware.txt
```

What Wazuh detected:



Skills Demonstrated

- Wazuh SIEM deployment and agent configuration
- Windows Event Log analysis
- Simulating brute force, persistence, and file drop attacks

- Configuring File Integrity Monitoring (FIM)
- Reading and filtering alerts in Wazuh dashboard
- Writing custom XML detection rules
- Using Wazuh search operators for log analysis